

Dupire PINN / PDF validation

PDF validation & IBP diagnostic: three case studies

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JCF manuscript — LocalVolatility PINN pipeline

December 2025 / May 2026

Outline

Overview

Synthetic: constant volatility ($\sigma = 1$)

Synthetic: manuscript Dupire volatility

Market data

IBP three-way diagnostic

Summary

Paper overview

Wang, Shaa, Privault, Guet — JCF manuscript

Two physics-informed neural networks calibrate Dupire local volatility:

- NN_phi: ansatz $\tilde{\varphi}(T, K) = 1 - e^{-\text{NN}_\varphi(T, K)}$ encodes call-price positivity and $K \rightarrow \infty$ decay
- NN_eta: local variance $\tilde{\eta}(T, K) = \sigma^2(T, K)$ fed directly into Dupire PDE

Three case studies:

1. Synthetic constant volatility $\sigma = 1$
2. Synthetic non-constant Dupire volatility (manuscript formula, §4.1)
3. Market data — DAX Aug 2001 / SPX May 2019

Validation: PDF analysis (MC histogram vs NN density) + IBP three-way mean diagnostic

Case 1 setup: constant volatility $\sigma = 1$

Model: $dS_t = rS_t dt + S_t dW_t$ ($r = 0.02$, $S_0 = 1000$)

Training data: synthetic option prices from Black–Scholes closed form

Strike range: $K \in [500, 3000]$

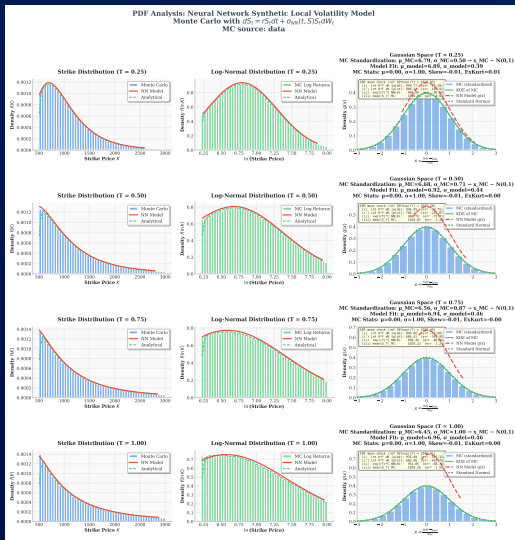
Maturities: $T \in \{0.5, 1.0, 1.5\}$

Benchmark: analytical BS density (log-normal)

Key quantities:

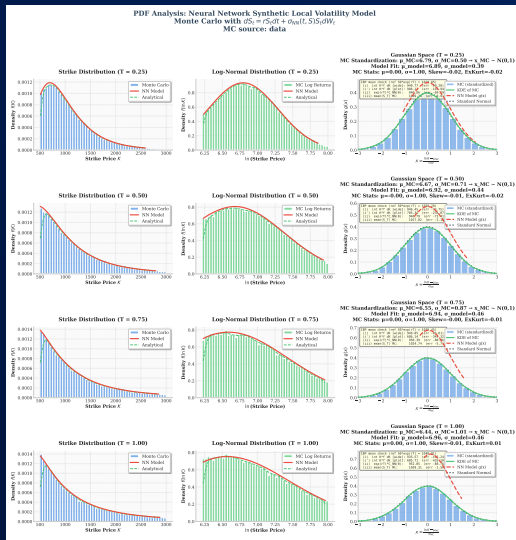
- Forward $S_0 e^{rT}$
- NN density $\partial^2 C_{\text{NN}} / \partial K^2$ vs MC histogram
- IBP mean
 $M_1 = e^{rT} \int K \partial^2 C / \partial K^2 dK$
- $M_2 = e^{rT} C_{\text{NN}}(0, T)$
- $M_3 = \mathbb{E}[S_T]$ from MC

Case 1: constant vol — PDF analysis (pretrained model)



MC source: data. Strike-space (left), log-strike (centre), Gaussian-standardized (right). IBP box

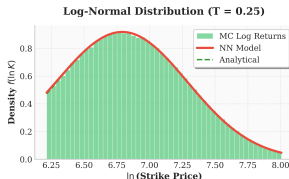
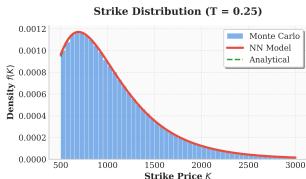
Case 1: constant vol — PDF analysis (repriced)



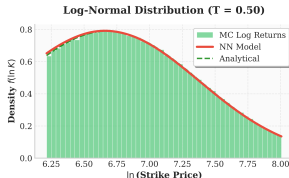
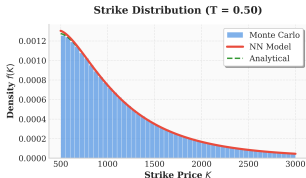
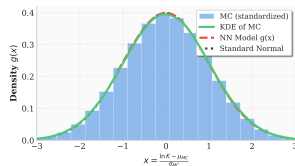
MC source: repriced. Same model; MC paths generated with the NN-implied $\sigma(t, S)$. M_3 now tracks

Case 1: constant vol $\sigma = 1$ — PDF analysis (Dec 9 2025)

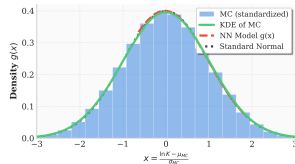
PDF Analysis: Neural Network Synthetic Local Volatility Model Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$



Gaussian Space (T = 0.25)
MC Standardization: $\mu_{MC}=6.79, \sigma_{MC}=0.50 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.41$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.00$



Gaussian Space (T = 0.50)
MC Standardization: $\mu_{MC}=6.68, \sigma_{MC}=0.71 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.92, \sigma_{model}=0.46$
MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.01, \text{ExKurt}=-0.01$



Gaussian Space (T = 0.75)
MC Standardization: $\mu_{MC}=6.56, \sigma_{MC}=0.87 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=6.94, \sigma_{model}=0.47$

Case 1: IBP mean diagnostics (pretrained model)

T	M_{ref}	M_1	%err	M_2	%err	M_3	%err	$\Delta M_{12}\%$	$\Delta M_{13}\%$
0.5	1020.20	948.87	-6.99	948.93	-6.99	1018.25	-0.19	0.006	6.814
1.0	1040.81	967.66	-7.03	967.74	-7.02	1038.19	-0.25	0.009	6.794
1.5	1061.84	982.49	-7.47	982.95	-7.43	1054.36	-0.70	0.047	6.816

- $M_1 \equiv M_2$ to $<0.05\%$ at all $T \Rightarrow$ IBP identity holds exactly in the code.
- M_3 within 0.7% of forward $\Rightarrow$ NN_η MC well-calibrated.
- $M_1, M_2 \approx 7\%$ below forward $\Rightarrow$ NN_φ undershoots at $K \rightarrow 0$.

Case 2 setup: non-constant Dupire volatility

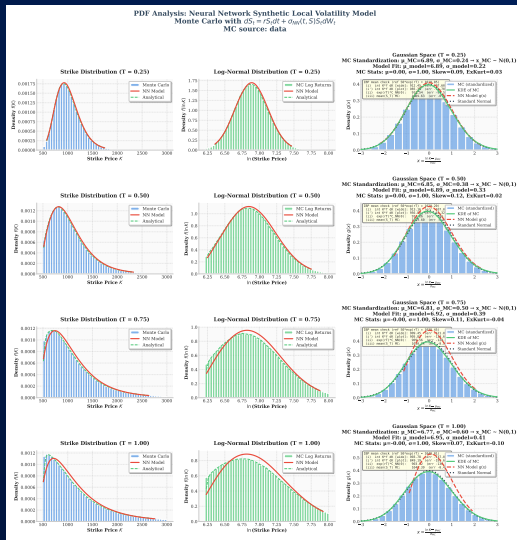
Manuscript formula (§4.1):

$$\sigma(x, t) = 0.3 + y e^{-y}, \quad y = (t + 0.1)\sqrt{x + 0.1}$$

where $x = S_t$, producing a non-trivial $\sigma^2(K, T)$ surface.

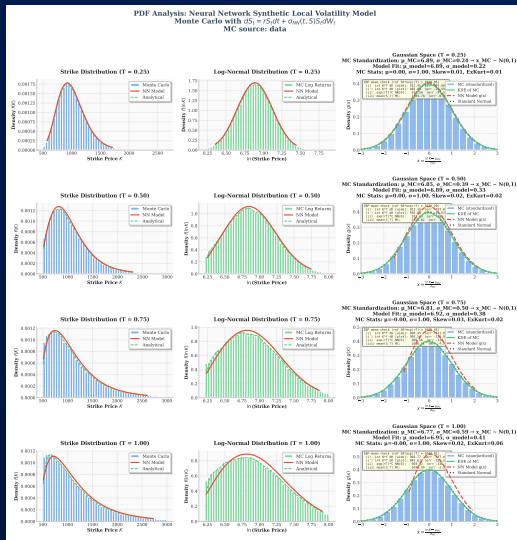
- $S_0 = 1000$, $r = 0.02$
- Strike range $K \in [500, 3000]$
- Maturities $T \in \{0.25, 0.50, 0.75, 1.00\}$
- Training data generated from the exact formula (not market quotes)
- Benchmark: analytical density from exact $\sigma(x, t)$
- NN must recover non-flat $\sigma^2(K, T)$ surface from prices alone
- IBP diagnostics: same three-way check as Case 1

Case 2: Dupire vol — PDF analysis (pretrained model)



MC source: data. Non-constant $\sigma(x, t)$ produces a skewed, non-log-normal density (visible in

Case 2: Dupire vol — PDF analysis (repriced)

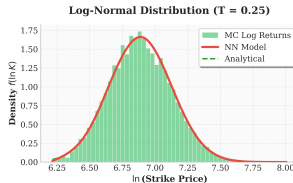
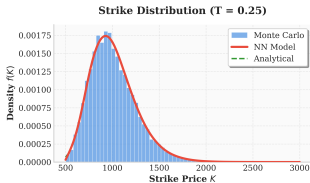


MC source: repriced. MC paths driven by NN-implied $\sigma(t, S)$. Shape match in Gaussian space

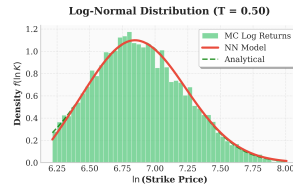
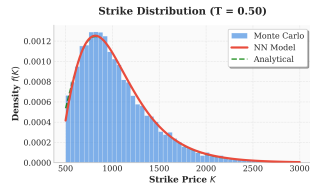
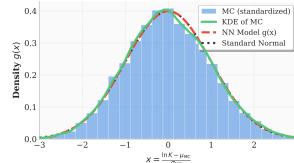
2025)

Case 2: Dupire non-constant vol — PDF analysis (Dec 7

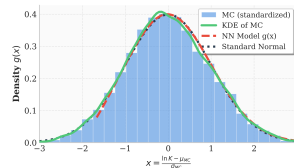
PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S) S_t dW_t$ 

Gaussian Space (T = 0.25)
 MC Standardization: $\mu_{MC}=6.89, \sigma_{MC}=0.24 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=6.89, \sigma_{model}=0.24$
 MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.08, \text{ExKurt}=-0.01$



Gaussian Space (T = 0.50)
 MC Standardization: $\mu_{MC}=6.85, \sigma_{MC}=0.38 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=6.90, \sigma_{model}=0.36$
 MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=0.09, \text{ExKurt}=-0.01$



Case 3: Market data — context

DAX Aug 7–9 2001: European index options, 3-day window.

- $K \in [4000, 9000]$, $T \in \{0.12, 0.20, 0.37, 0.60, 0.87\}$
- NN trained on DAX market quotes; MC driven by NN-implied $\sigma(t, S)$
- $\mu_{MC} \approx 8.64\text{--}8.66$ (in $\ln K$ space)
- NN model deviates noticeably from MC/analytical in K -space (visible panel 1)
- Gaussian-space MC tracks $\mathcal{N}(0, 1)$; NN $g(x)$ deviates — $K=0$ boundary fingerprint

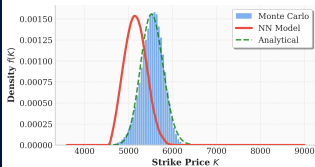
Provisional day-to-file assignment inferred from chronological filename order. **Confirm via raw data** if day ordering matters for interpretation.

Case 3: DAX Aug 7, 2001 — PDF analysis

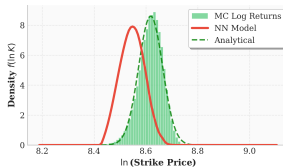
PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$

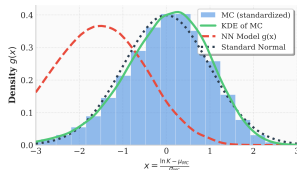
Strike Distribution (T = 0.12)



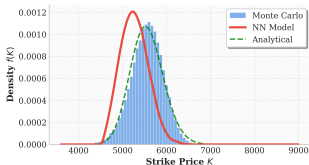
Log-Normal Distribution (T = 0.12)



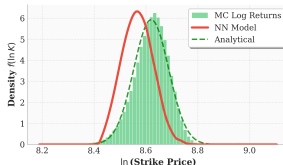
Gaussian Space (T = 0.12)
MC Standardization: $\mu_{MC}=8.62, \sigma_{MC}=0.05 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=8.55, \sigma_{model}=0.05$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.46, \text{ExKurt}=0.52$



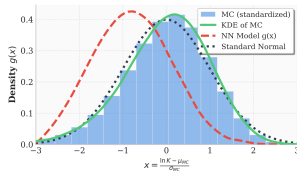
Strike Distribution (T = 0.20)



Log-Normal Distribution (T = 0.20)



Gaussian Space (T = 0.20)
MC Standardization: $\mu_{MC}=8.62, \sigma_{MC}=0.07 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=8.57, \sigma_{model}=0.06$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.50, \text{ExKurt}=0.77$



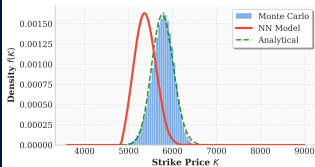
Gaussian Space (T = 0.37)
MC Standardization: $\mu_{MC}=8.62, \sigma_{MC}=0.10 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=8.60, \sigma_{model}=0.08$

Case 3: DAX Aug 8, 2001 — PDF analysis

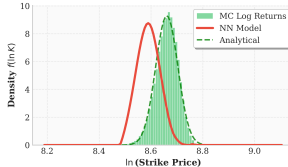
PDF Analysis: Neural Network Synthetic Local Volatility Model

Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$

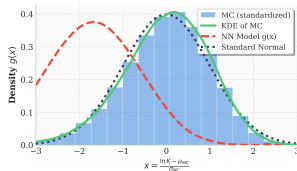
Strike Distribution (T = 0.12)



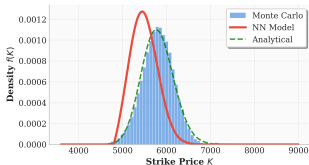
Log-Normal Distribution (T = 0.12)



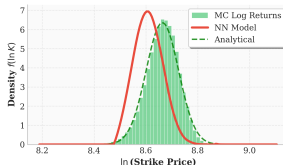
Gaussian Space (T = 0.12)
 MC Standardization: $\mu_{MC}=8.66, \sigma_{MC}=0.04 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=8.59, \sigma_{model}=0.04$
 MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.41, \text{ExKurt}=0.47$



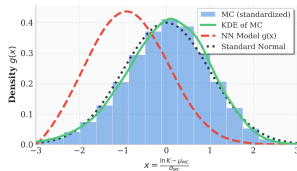
Strike Distribution (T = 0.20)



Log-Normal Distribution (T = 0.20)



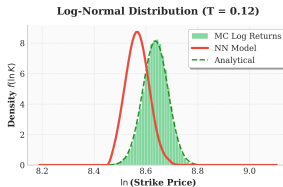
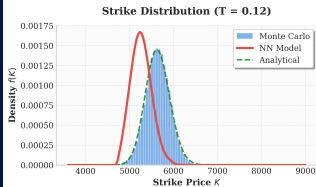
Gaussian Space (T = 0.20)
 MC Standardization: $\mu_{MC}=8.66, \sigma_{MC}=0.06 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=8.61, \sigma_{model}=0.06$
 MC Stats: $\mu=0.00, \sigma=1.00, \text{Skew}=-0.43, \text{ExKurt}=0.71$



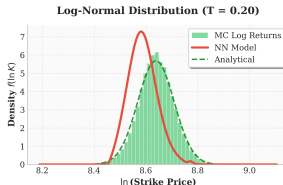
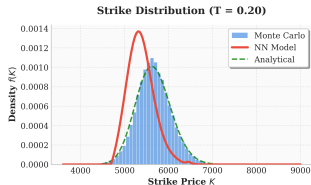
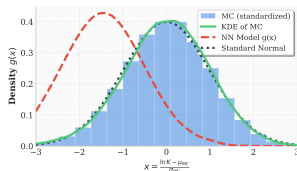
Gaussian Space (T = 0.37)
 MC Standardization: $\mu_{MC}=8.67, \sigma_{MC}=0.09 \rightarrow x_{MC} \sim N(0,1)$
 Model Fit: $\mu_{model}=8.65, \sigma_{model}=0.07$

Case 3: DAX Aug 9, 2001 — PDF analysis

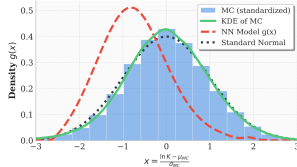
PDF Analysis: Neural Network Synthetic Local Volatility Model
Monte Carlo with $dS_t = rS_t dt + \sigma_{NN}(t, S)S_t dW_t$



Gaussian Space (T = 0.12)
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Model Fit: $\mu_{model}=8.57, \sigma_{model}=0.04$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.23, \text{ExKurt}=0.46$



Gaussian Space (T = 0.20)
MC Standardization: $\mu_{MC}=8.64, \sigma_{MC}=0.07 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=8.59, \sigma_{model}=0.06$
MC Stats: $\mu=-0.00, \sigma=1.00, \text{Skew}=-0.25, \text{ExKurt}=0.98$



Gaussian Space (T = 0.37)
MC Standardization: $\mu_{MC}=8.64, \sigma_{MC}=0.10 \rightarrow x_{MC} \sim N(0,1)$
Model Fit: $\mu_{model}=8.62, \sigma_{model}=0.07$

IBP three-way diagnostic: M_1, M_2, M_3

For each maturity T , compute three independent estimates of $\mathbb{E}[S_T]$ from the trained NN pair $(\text{NN}_\varphi, \text{NN}_\eta)$:

$$M_1 = e^{rT} \int_0^\infty K \frac{\partial^2 C_{\text{NN}}}{\partial K^2} dK \quad (\text{density quadrature})$$

$$M_2 = e^{rT} C_{\text{NN}}(0, T) \quad (\text{IBP closed form})$$

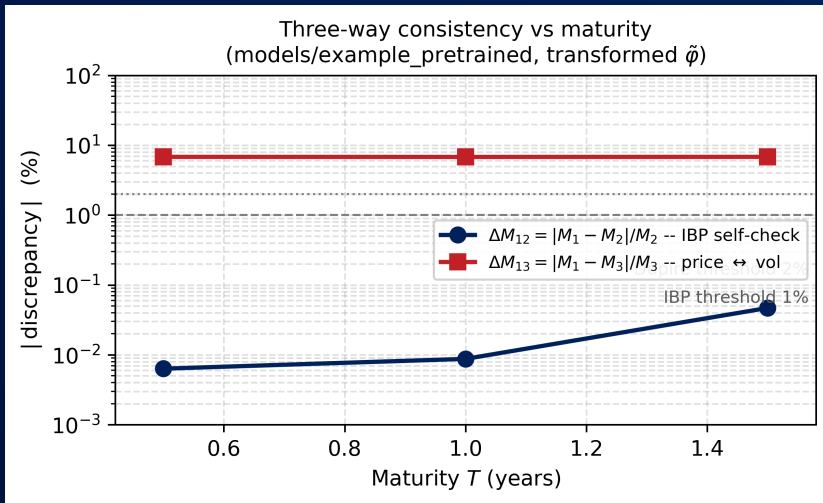
$$M_3 = \mathbb{E}[S_T] \text{ from MC with } \sigma_{\text{NN}}(t, S) \quad (\text{Dupire forward simulation})$$

Under no-arbitrage all three should equal $S_0 e^{rT}$.

T	M_{ref}	M_1	%err	M_2	%err	M_3	%err	$\Delta M_{12}\%$	$\Delta M_{13}\%$
0.5	1020.20	948.87	-6.99	948.93	-6.99	1018.25	-0.19	0.006	6.814
1.0	1040.81	967.66	-7.03	967.74	-7.02	1038.19	-0.25	0.009	6.794
1.5	1061.84	982.49	-7.47	982.95	-7.43	1054.36	-0.70	0.047	6.816

Pretrained models/example-pretrained on the manuscript Dupire surface.

Discrepancy: ΔM_{12} and ΔM_{13} vs maturity



- $\Delta M_{12} \equiv |M_1 - M_2|/M_2 < 0.05\%$ everywhere $\Rightarrow$ IBP identity holds to machine precision; autodiff and quadrature agree.

Hypothesis: the $K=0$ boundary is structurally unreachable

Implemented call-price ansatz (as trained, `neural_phi_tilde`):

$$\pi_{\theta}^c(k, t) = S_0(1 - e^{-\mathcal{N}_c(k, t; \theta_c)}), \quad \mathcal{N}_c = \text{softplus output} > 0.$$

So $\tilde{\varphi} = 1 - e^{-\mathcal{N}_c}$ has range $(0, 1)$ — **open at 1**.

- $K = \infty$ ($k = 1$): BC is $\tilde{\varphi} = 0$, i.e. $\mathcal{N}_c = 0$ — **attainable**.
- $K = 0$ ($k = 0$): BC is $\tilde{\varphi} = 1$, i.e. $\mathcal{N}_c \rightarrow \infty$ — **unattainable** by a finite output. Hence $C_{\text{NN}}(0, T) < S_0$ by construction of the map, and the gap is our 7%.

(Eq. 3.9 of the manuscript carries an extra $(1 - k)$ factor that is *not* in the code. It would not help here: at $k = 0$, $(1 - k) = 1$, so the $K = 0$ deficit persists either way.)

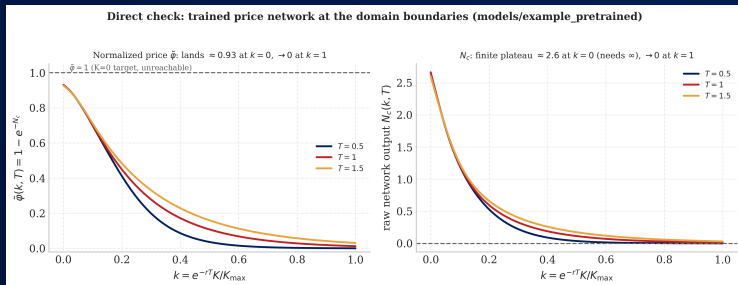
Why the deficit is constant in T : the rescaled Dupire PDE $\partial_t \pi = \eta k^2 \partial_k^2 \pi$ carries a k^2 factor that vanishes at $k = 0$, so $\pi_{\theta}^c(0, t) = \pi_{\theta}^c(0, 0)$ for all t : the $T = 0$ deficit is frozen across every maturity — exactly the flat $\Delta M_{13} \approx 7\%$.

Direct check: evaluate $\mathcal{N}_c$ at the boundaries

Evaluate the trained network at $k = 0$ and $k = 1$ (models/example_pretrained):

T	$\mathcal{N}_c(0, T)$	$\tilde{\varphi}(0, T)$	deficit	$\mathcal{N}_c(1, T)$	$\tilde{\varphi}(1, T)$
0.5	2.661	0.9301	6.99%	0.0007	0.0007
1.0	2.656	0.9298	7.02%	0.0128	0.0128
1.5	2.600	0.9257	7.43%	0.0315	0.0310

$\mathcal{N}_c(0, T) \approx 2.6$ (finite — would need ∞) vs. $\mathcal{N}_c(1, T) \approx 0.01$: two orders of magnitude. The far boundary is nearly reached; $K=0$ is not. $e^{rT} C_{\text{NN}}(0, T)$ reproduces $M_2 = 948.9/967.7/983.0$.



Summary

- Across constant- σ , manuscript Dupire, and DAX market data, the published NN pair fits the in-window grid well.
- The IBP three-way test reveals a T -independent $\sim 7\%$ gap between $M_1 = M_2$ and M_3 .
- ΔM_{12} is at machine precision; M_3 within 0.7% of forward. The gap is entirely in NN_φ , not NN_η .
- Root cause: the output map $\tilde{\varphi} = 1 - e^{-\mathcal{N}_c}$ has range $(0, 1)$. The $K = \infty$ BC ($\tilde{\varphi} = 0$) is reachable ($\mathcal{N}_c = 0$); the $K = 0$ BC ($\tilde{\varphi} = 1$) needs $\mathcal{N}_c \rightarrow \infty$ and is not. The Dupire PDE degenerates at $k = 0$, freezing the deficit across T . Confirmed directly: $\mathcal{N}_c(0, T) \approx 2.6$, $\mathcal{N}_c(1, T) \approx 0.01$.